Micro-Q-Instruct

A Degradation-Aware Microscopy Benchmark: From Controlled Classical Image Processing to Paired Vision-Language Model Robustness

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Abstract

Vision-language models (VLMs) increasingly encounter microscopy images through acquisition pipelines that introduce blur, noise, sub-sampling, compression, and detector artifacts. A model's semantic answer about a specimen may depend on this low-level acquisition quality just as much as on biological content, yet natural-image quality benchmarks rarely model microscopy physics, and microscopy-focused VLM benchmarks rarely isolate low-level corruption. This project, *Micro-Q-Instruct*, addresses that gap with a traceable, paired benchmark in which every corruption and restoration step is known and reproducible. Starting from 1,701 clean microscopy sources drawn from permissive training/development pools and the pinned, official μ-Bench evaluation set, we generate 15,480 degradation recipes spanning ten physically motivated degradation families at three severity levels, producing 30,960 paired derivatives (degraded and restored). Each image is characterized by a 19-feature classical fingerprint and scored with paired fidelity metrics (MSE, PSNR, SSIM, edge retention, histogram divergence). A matched classical restoration strategy is applied per family, and 99.95% of eligible severity triplets are verified to be PSNR-monotonic. In a second phase, we stress-test Qwen3-VL-8B on 500 unique paired Micro-Bench samples across clean, degraded, and restored image conditions while holding the question, options, model, prompt, and decoding parameters fixed. We find that restoration recovers most, but not all, of the aggregate semantic robustness loss (66.8% $\rightarrow$ 65.2% $\rightarrow$ 66.4% macro accuracy), that recovery is strongly family-dependent (periodic interference: -12.9 pp degradation loss, $+11.5$ pp restoration gain; photobleaching and quantization remain information-limited), and that fine-grained perception tasks are the primary stress point under strong corruption. The result is a reproducible, audit-tracked classical DIP benchmark coupled with an inference-time robustness study that connects pixel-level image quality to downstream VLM correctness.

Contents

1 Introduction and Motivation **3**

1.1 Research Questions 3

1.2 Course Alignment 3

2 Project Overview: A Two-Phase Causal Chain **3**

3 Phase 1: Data Sourcing and Integrity **4**

3.1 Strict Two-Pool Policy 4

3.2 Balanced, Deterministic Degradation Planning 5

4 Phase 1: The Ten Degradation Families **5**

5 Phase 1: Measurement and Restoration Framework **5**

5.1 19-Feature Interpretable Fingerprint 5

5.2 Paired Image Evidence 5

5.3 Restoration Strategy Registry 6

5.4 Release Validation 7

6 Phase 1: Results **7**

6.1 Feature Space Heterogeneity 7

6.2 Paired Fidelity Distributions 7

6.3 Restoration Is Governed by Invertibility, Not Visual Polish 7

6.4 Severity Is Empirically Ordered but Not Equally Spaced 8

7 Phase 1.5: Paired Inference-Time Robustness **8**

7.1 Experimental Design 8

7.2 Implementation 8

7.3 Statistical Methodology 9

8 Phase 1.5: Results **9**

8.1 Overall Robustness 9

8.2 Degradation-Specific Response 9

8.3 Severity Response 10

8.4 Paired Correctness Transitions 10

9 Discussion and Conclusions **11**

1 Introduction and Motivation

Vision-language models are increasingly used to interpret microscopy images for tasks ranging from coarse modality recognition to fine-grained morphological reasoning. However, a VLM does not see the specimen directly — it sees the specimen *through an acquisition pipeline*. A correct semantic answer may depend on blur, noise, sampling, compression, registration, and detector response, not solely on biological content.

This motivates the central question addressed by this project: *does image quality belong in microscopy reasoning benchmarks, and if so, how should it be measured?* Two lines of prior work are relevant but insufficient on their own:

- Natural-image quality benchmarks (e.g., low-level vision curricula such as Q-Instruct and Q-Bench) rarely model microscopy-specific physics such as Poisson-Gaussian shot noise, photobleaching, or PSF-governed defocus.
- Microscopy-focused VLM benchmarks (e.g., μ-Bench) evaluate semantic and biological reasoning but rarely isolate low-level image corruption as an independent, controlled variable.

Our response is a **traceable, paired benchmark** in which every corruption and every restoration is known: exact parameters, random seeds, bit depth, content hashes, and source licenses are recorded for every generated image, so quality $\rightarrow$ evidence $\rightarrow$ answer can be audited end to end.

1.1 Research Questions

The project is organized around three research questions, all evaluated on a shared, paired experimental spine in which *only the pixels change* — the question, answer options, correct answer, model, prompt template, and decoding parameters are held fixed:

1. **Detect.** Can an interpretable digital image processing (DIP) fingerprint identify degradation family and severity across heterogeneous microscopy domains (fluorescence, light, electron)?
2. **Restore.** Which matched classical restoration method recovers fidelity, edge content, and morphology for each degradation family — and where is information irreversibly lost?
3. **Stress-test.** Does a modern VLM (Qwen3-VL) preserve the same correct answer across clean, degraded, and restored views of the same image?

1.2 Course Alignment

The project was designed so that essentially every chapter of the Digital Image Processing syllabus becomes an implemented, measurable artifact rather than a purely theoretical exercise. Table 1 summarizes this mapping.

2 Project Overview: A Two-Phase Causal Chain

The pipeline is structured as a causal chain that moves from raw microscopy images to robustness evidence:

Chapter	Topic	Implemented artifact
Ch. 2	Digital fundamentals	Sampling, anti-aliasing, quantization, pixel precision
Ch. 3	Spatial processing	PSF modeling, gradients, Laplacian, gamma, CLAHE
Ch. 4	Frequency domain	Spectral energy, sinusoidal interference, notch rejection
Ch. 5	Restoration	Poisson/read noise modeling, local Wiener filtering, deconvolution
Ch. 7	Transforms	Wavelet energy fingerprints and denoising
Ch. 8	Compression	Controlled JPEG and block-boundary evidence
Ch. 9	Morphology	Otsu proxy masks, Dice/IoU, connected components
Bridge	Vision-language robustness	Paired inference, uncertainty, exact-answer scoring

Table 1: Mapping of DIP course chapters to implemented components of the benchmark. Foundational references: μ-Bench (NeurIPS 2024), Q-Instruct (CVPR 2024), Q-Bench (ICLR 2024).

Clean microscopy $\rightarrow$ Controlled degradation $\rightarrow$ DIP fingerprint $\rightarrow$ Matched restoration $\rightarrow$ Paired VLM test

Phase 1 — Classical DIP benchmark. Produces paired images, exact generation recipes, interpretable per-image measurements, morphology metrics, restored images, and a classical restoration baseline. This phase involves no learned model and is fully deterministic given a master random seed.

Phase 1.5 — Inference-time robustness. Measures answer stability of a vision-language model (Qwen3-VL) with paired statistics across the clean / degraded / restored conditions. No model training or fine-tuning occurs in this phase; it is a pure evaluation study.

3 Phase 1: Data Sourcing and Integrity

3.1 Strict Two-Pool Policy

To avoid any contamination between data used for calibrating the pipeline and data used for reporting final numbers, the project enforces a strict train/evaluate separation: classical strategies and thresholds are tuned on permissive external sources, while all reported evaluation numbers come exclusively from the pinned, official μ-Bench evaluation set.

From 1,701 unique clean source images, the pipeline allocates:

- **804** images to *training* (external sources such as BBBC010, BBBC038, IDR0067);
- **147** images to *development*, held out purely for calibration;
- **750** images to *evaluation*, drawn from 15 official μ-Bench datasets, with the μ-Bench revision pinned to commit `3f2c5b59...`

The modality composition of the pool is fluorescence-heavy but multi-modal: 893 fluorescence, 625 light microscopy, and 183 electron microscopy images. All records retain their original license (CC0, CC0-1.0, CC-BY-4.0, or CC-BY-SA-4.0). A **contamination guard** combines decoded-pixel SHA-256 hashing, a 64-bit perceptual-hash nearest-neighbor audit, and a license allowlist to ensure no evaluation image leaks into the training or development pools.

3.2 Balanced, Deterministic Degradation Planning

Rather than applying every degradation family to every image (a full factorial design), the pipeline uses a **balanced incomplete factorial**: for each clean source, nine variants are generated by selecting three physically applicable degradation families and combining each with weak, medium, and strong severity. Deterministic clean controls and a configured fraction of mixed (stacked) corruptions are added on top of this design. The entire generation process is seeded with a single master seed (`20260724`), making every recipe exactly reproducible.

Applicability is **physics-aware**: for example, photobleaching and channel-shift degradations are restricted to fluorescence images; chromatic misregistration requires genuinely multichannel content; and defocus blur is restricted to optical (light and fluorescence) modalities. Degradation parameters additionally scale with image size so that severity is comparable across differently sized sources.

This design yields:

- **10** degradation families,
- **3** severity levels (weak / medium / strong),
- **15,480** degraded images,
- **15,480** matched restored images,
- **30,960** total paired derivatives.

Every recipe preserves full provenance: image role (train/dev/eval), exact degradation and restoration parameters, random seed, bit depth, SHA-256 content hash, source license, and output path.

4 Phase 1: The Ten Degradation Families

Each degradation family is implemented as a physically motivated forward model with a matched classical restoration strategy. Representative results at **strong** severity are summarized below, with paired image evidence (PSNR, SSIM, and their restoration deltas ΔPSNR, ΔSSIM).

5 Phase 1: Measurement and Restoration Framework

5.1 19-Feature Interpretable Fingerprint

Every image (clean, degraded, and restored) is characterized by a 19-dimensional classical feature vector, deliberately chosen to be interpretable rather than learned: histogram statistics, entropy, gradient and Laplacian energy, Fourier band-energy ratios, spectral centroid, wavelet sub-band energy, blockiness, clipping fraction, and inter-channel shift.

5.2 Paired Image Evidence

For every degraded/restored pair relative to its clean reference, the pipeline computes:

Family	Degradation model	Matched restoration
1. Sampling / resolution loss	Downsample then reconstruct at original dimensions, removing spatial samples and exposing aliasing/blur.	BayesShrink db2 wavelet denoising (artifact suppression only; removed samples are not invertible).
2. Bit-depth quantization	Uniform intensity quantization collapses the continuous tonal range into a small set of discrete levels.	Wavelet denoising softens contouring but cannot recreate discarded intensity levels.
3. Poisson–Gaussian noise	Photon-count shot noise combined with additive read noise to mimic noisy acquisition.	Local adaptive Wiener filter using neighborhood statistics to attenuate signal-dependent noise.
4. Defocus blur	Normalized disk point-spread function (PSF) removes high-frequency focus cues.	Recorded-PSF Wiener deconvolution, trading sharpness against ringing/noise amplification.
5. Motion / scan drift	Normalized linear PSF smears structures along a controlled length and angle.	Recorded linear-PSF Wiener deconvolution, partially reconstructing edge direction and texture.
6. Periodic interference	Oriented sinusoid adds coherent striping at a known spatial frequency and amplitude.	Paired notch-reject filter removing the synthetic frequency peaks (an explicitly oracle-assisted baseline).
7. JPEG compression	Lossy block-transform coding discards coefficients, introducing block/ringing artifacts; 16-bit inputs also incur an 8-bit round trip.	Wavelet denoising smooths post-compression artifacts, but discarded coefficients remain unrecoverable.
8. Photobleaching	Fluorescence signal is attenuated above background (models retained intensity, not detector clipping).	Gamma correction plus CLAHE restores visibility/local contrast without recreating lost photons.
9. Chromatic misregistration	Opposing subpixel channel shifts create color fringes in multichannel fluorescence images.	Phase-correlation channel registration estimates displacement relative to the green channel and realigns red/blue.
10. Detector saturation	Gain and black-level changes followed by detector-range clipping, flattening bright structures.	Gamma/CLAHE enhancement redistributes visible contrast; clipped values cannot be reconstructed.

Table 2: The ten degradation families, their forward (degradation) models, and their matched classical restoration strategies.

- pixel-fidelity metrics: MSE, PSNR, SSIM;
- structural metrics: edge-retention ratio, histogram Jensen–Shannon divergence;
- where segmentation masks exist: Dice coefficient, IoU, region area, and connected-component counts (via Otsu proxy masks).

5.3 Restoration Strategy Registry

A registry of six classical restoration strategies is applied per family according to the matching in Table 2: local Wiener filtering, PSF-aware deconvolution, frequency-domain notch rejection, phase-correlation channel registration, gamma correction with CLAHE, and wavelet-domain comparison/denoising.

5.4 Release Validation

The full Phase 1 release comprises **15,480** materialized recipes generated from **1,701** clean source images, split into **7,327** training, **6,822** evaluation, and **1,331** development records. Validation of the release reports **zero errors and zero warnings** across all 15,480 analyses, and **3,743 of 3,745** eligible severity triplets (99.95%) satisfy strict PSNR monotonicity, i.e. $\mathrm{PSNR}(\text{weak}) \geq \mathrm{PSNR}(\text{medium}) \geq \mathrm{PSNR}(\text{strong})$. The release package includes recipe / generated / restored JSONL manifests, paired CSV and JSONL analysis records, a dataset card, an audit log, validation and calibration figures, and serialized classical baselines.

6 Phase 1: Results

6.1 Feature Space Heterogeneity

Exploratory analysis of the 19-feature fingerprint across the full dataset shows that microscopy images span radically different visual regimes. The entropy distribution is broad and multimodal, with low-information fields (e.g. sparse fluorescence puncta) coexisting with richly textured brightfield specimens, so a single global threshold for any detector would be brittle. Intensity variation (feature.intensity_std) is strongly right-skewed: most images have modest contrast, with a small high-contrast tail that motivates per-modality normalization rather than a single global scale.

6.2 Paired Fidelity Distributions

Aggregate PSNR values concentrate in the 25–40 dB range but retain long tails in both directions, reflecting the coexistence of subtle clean-control cases and destructive strong corruptions. SSIM piles up near 1.0 while still spanning the full $[0, 1]$ range, confirming that PSNR and SSIM capture complementary failure modes and that neither scalar alone summarizes degradation impact.

6.3 Restoration Is Governed by Invertibility, Not Visual Polish

Table 3 reports the mean evaluation-set restoration delta for representative single-degradation families.

Family	ΔPSNR (dB)	ΔSSIM	Interpretation
Periodic interference	$+7.79$	$+0.109$	known-frequency notch filter excels
Poisson–Gaussian noise	$+5.21$	$+0.165$	local statistics are recoverable
Motion / scan drift	$+1.43$	$+0.044$	only partial PSF inversion possible
Bit-depth quantization	-2.59	-0.038	discarded levels cannot be regenerated
Photobleaching	-12.11	-0.274	contrast enhancement $\neq$ photon recovery

Table 3: Mean evaluation-set ΔPSNR / ΔSSIM after restoration for selected single-degradation families. Positive values indicate the restoration improves fidelity relative to the degraded image; negative values indicate that classical enhancement further reduces measured fidelity even though it may subjectively improve visibility.

The key finding is that **restoration effectiveness tracks the invertibility of the underlying forward model**, not the perceptual quality of the output. Degradations with a known, structured forward operator (periodic interference, additive signal-dependent noise, and to a lesser extent linear motion blur) are substantially

recoverable. Degradations that destroy information irreversibly — discarded quantization levels, clipped detector saturation, and attenuated fluorescence signal — cannot be recovered by contrast enhancement alone, and enhancement can even *reduce* measured pixel fidelity while improving apparent visibility.

As a secondary analysis, a simple cross-source linear classifier over the 19-feature fingerprint was evaluated as a baseline degradation detector. It achieves 34.9% degradation-family accuracy (macro-F1 27.3%) and 41.0% severity accuracy (macro-F1 38.2%). This confirms that classical features carry transferable signal about degradation type and strength, but that cross-modality and cross-domain variation prevents a simple linear fingerprint from becoming a universal, high-accuracy detector — motivating the more targeted, per-family restoration approach used elsewhere in the pipeline.

6.4 Severity Is Empirically Ordered but Not Equally Spaced

Across all ten degradation families, 99.95% of eligible paired severity triplets satisfy $\mathrm{PSNR}(\text{weak}) \geq \mathrm{PSNR}(\text{medium}) \geq \mathrm{PSNR}(\text{strong})$, confirming that the severity parameterization behaves monotonically as intended. However, the weak→medium→strong slopes differ substantially by family (e.g. chromatic misregistration and sampling loss degrade gently, while quantization, Poisson–Gaussian noise, and detector saturation degrade steeply), showing that severity must be calibrated per corruption type rather than treated as a single universal physical scale.

7 Phase 1.5: Paired Inference-Time Robustness

7.1 Experimental Design

Phase 1.5 asks whether a vision-language model's semantic correctness on the official μ-Bench evaluation questions is preserved across the same three pixel conditions used in Phase 1. For each unique sample (an evaluation image, its official multiple-choice question, and one Phase-1 degradation recipe), three pixel conditions are constructed:

- **Clean** — the official μ-Bench image, unmodified;
- **Degraded** — the exact Phase 1 recipe applied to that image;
- **Restored** — the matched restored image from Phase 1.

Held constant across all three conditions: the question text, answer options, expected answer, prompt template, model, decoding parameters, and a sample-specific random seed. **Changed deliberately**: only the image pixels. Prompts never reveal which condition (clean, degraded, restored) is being shown, and request order is deterministically shuffled to reduce order and load effects. The unit of analysis is the independent image–question triplet; there is no pseudoreplication from asking multiple questions about the same source image.

7.2 Implementation

The evaluation pipeline joins recipe, generated-image, restored-image, and official annotation records by recipe ID; balances the sampled cohort across task type, degradation family, severity, and modality (one

question and one recipe per image); and interleaves all three conditions in a deterministic, shuffled request order. Inference is performed with **Qwen3-VL-8B** served via vLLM, using deterministic decoding and a structured JSON answer request. Responses are parsed strictly by JSON field, answer letter, index, or exact option text — ambiguous responses are never guessed as correct.

The resulting cohort comprises **500** unique paired samples (167 coarse-perception and 333 fine-perception questions; 100 electron, 100 fluorescence, and 300 light microscopy samples), yielding **1,500** completed inference requests across **6** perception task types, with a **100%** answer-parse rate.

7.3 Statistical Methodology

The primary endpoint is **task-macro accuracy**,

$$\mathrm{MacroAcc}(c) \;=\; \mathrm{mean}_{\text{task}}\; \mathrm{mean}_{\text{sample}}\; \mathbb{1}[\hat{y}_{i,c} = y_i]\,,$$

where each of the six task types receives equal weight so that frequent task types cannot dominate the aggregate score. Uncertainty is quantified with a 10,000-resample paired bootstrap over complete image–question triplets, reporting 95% percentile confidence intervals. Condition-to-condition differences are additionally tested with the exact McNemar test on discordant pairs, with Holm correction across the three pairwise condition comparisons. Four paired effects are reported: degradation loss (clean $\rightarrow$ degraded), restoration gain (degraded $\rightarrow$ restored), residual versus clean (restored $\rightarrow$ clean), and recovery fraction. A full per-sample transition audit classifies each triplet into one of eight categories (e.g. stable correct, restored after failure, unrecovered failure, harmed by restoration).

It is important to note the **scope boundary** of this phase: the completed Phase 1 annotations only support coarse- and fine-perception question types; no higher-level cognition category is inferred or fabricated for this analysis.

8 Phase 1.5: Results

8.1 Overall Robustness

Overall macro accuracy moves from **66.8%** (clean) to **65.2%** (degraded) to **66.4%** (restored) — a -1.62 percentage-point (pp) degradation effect followed by a $+1.21$ pp restoration gain. All three condition-level 95% confidence intervals overlap, so the aggregate effect is directionally consistent but not individually significant at $n = 500$.

Splitting by question granularity reveals a more informative pattern: **coarse perception** accuracy (77.2% $\rightarrow$ 78.4% $\rightarrow$ 79.0%) shows no consistent loss and, if anything, a slight positive trend, whereas **fine perception** accuracy (61.6% $\rightarrow$ 58.6% $\rightarrow$ 60.1%) drops by **3.02 pp** under degradation and recovers only **1.51 pp** with restoration — confirming that fine-grained perceptual judgments are considerably more sensitive to acquisition-level image quality than coarse, modality-level judgments.

8.2 Degradation-Specific Response

Aggregate averages conceal substantial heterogeneity across degradation families (Table 4). **Periodic interference** produces by far the largest robustness loss (-12.9 pp, clean$\rightarrow$degraded) but is also nearly fully recovered by restoration ($+11.5$ pp, degraded$\rightarrow$restored), leaving only a -1.4 pp residual relative to clean

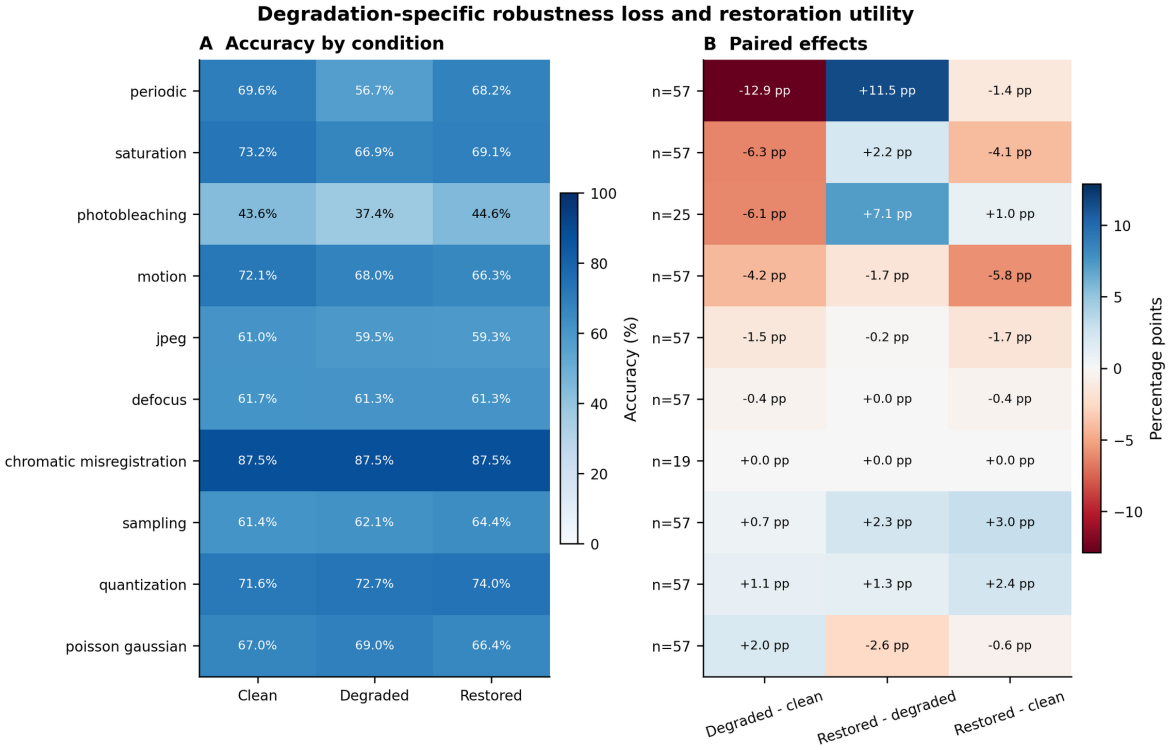


Figure 1: Showing degradation specific robustness and restoration effect on measured accuracies

— mirroring the pixel-level finding that structured, known-frequency corruption is highly invertible. **Photobleaching** shows a smaller degradation loss (-6.1 pp) but a comparatively large restoration gain ($+7.1$ pp), i.e. gamma/CLAHE contrast enhancement helps the VLM even though it does not fully restore pixel fidelity. By contrast, **motion blur** restoration is actually *harmful* at the semantic level (-1.7 pp degraded→restored, widening the residual to -5.8 pp versus clean), illustrating that a pixel-level restoration success does not guarantee a downstream semantic benefit. **Chromatic misregistration** shows no measurable effect in either direction, and subgroup sample sizes range from $n = 19$ to $n = 57$, so family-specific point estimates should be read with appropriate caution.

8.3 Severity Response

Semantic robustness separates most clearly under **strong** corruption: clean accuracy of 66.6% falls to 60.6% under strong degradation (-6.0 pp), and restoration recovers only $+2.0$ pp, leaving a -4.0 pp residual relative to clean. In contrast, weak and medium severity strata show no consistent accuracy loss, with wide, overlapping confidence bands. This indicates that the semantic robustness of the model is largely preserved at low corruption levels and degrades primarily once corruption becomes severe.

8.4 Paired Correctness Transitions

A per-sample transition audit across all 500 unique Micro-Bench samples shows that **59.6%** (298/500) of samples remain correct across all three conditions (*stable correct*), while **27.2%** (136/500) are incorrect under every condition (*always incorrect*). Restoration **fixes** 13 samples that were wrong under degradation,

Family	Clean %	Degraded %	Restored %	Deg.−Clean (pp)	Rest.−Deg. (pp)
Periodic interference	69.6	56.7	68.2	−12.9	+11.5
Detector saturation	73.2	66.9	69.1	−6.3	+2.2
Photobleaching	43.6	37.4	44.6	−6.1	+7.1
Motion / scan drift	72.1	68.0	66.3	−4.2	−1.7
JPEG compression	61.0	59.5	59.3	−1.5	−0.2
Defocus blur	61.7	61.3	61.3	−0.4	+0.0
Chromatic misregistration	87.5	87.5	87.5	0.0	0.0
Sampling / resolution loss	61.4	62.1	64.4	+0.7	+2.3
Bit-depth quantization	71.6	72.7	74.0	+1.1	+1.3
Poisson–Gaussian noise	67.0	69.0	66.4	+2.0	−2.6

Table 4: Qwen3-VL macro accuracy by degradation family and pixel condition, subgroup n ranges 19–57. pp = percentage points.

misses 16 samples that remain wrong after restoration, and actively **harms** 7 samples that were correct under degradation but become incorrect after restoration. Splitting by granularity, coarse-perception questions are stable correct 74% of the time (17% always incorrect), while fine-perception questions are stable correct only 53% of the time (32% always incorrect) — reinforcing that fine perception is the primary locus of semantic fragility. A per-family recovery map further shows that periodic interference lies close to the line of full recovery (robustness loss ≈ restoration gain), whereas motion blur and Poisson–Gaussian noise lie *below* that line, indicating restoration does not fully offset — and in the case of motion blur, compounds — the semantic damage.

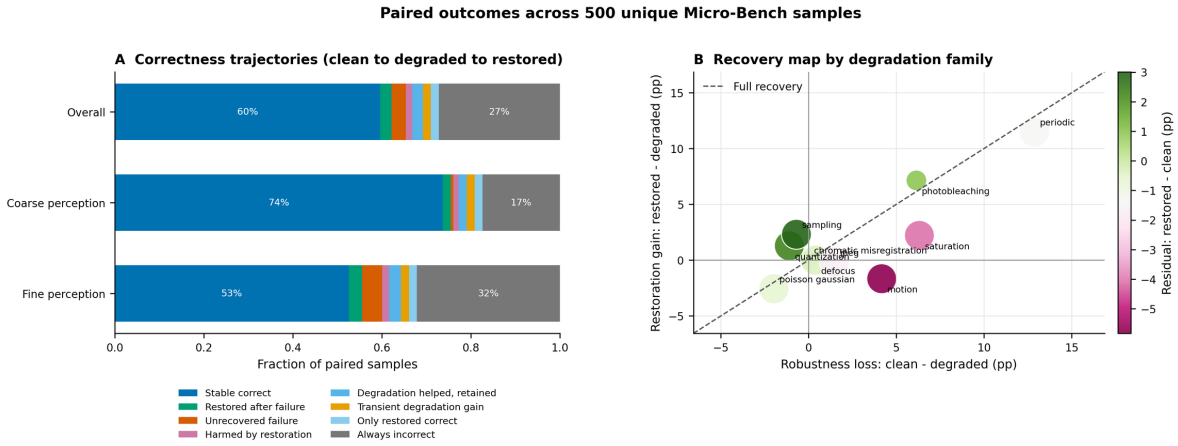


Figure 2: Shows a correctness trajectories regarding how restoration and degradation effected the predicted label. Also recovery map by degradation family is shown here as well.

9 Discussion and Conclusions

Three main conclusions are supported by the combined evidence from Phase 1 and Phase 1.5:

1. **A reproducible classical DIP benchmark.** Starting from 1,701 clean sources, the pipeline produces 15,480 fully provenanced degradation/restoration recipes with strict train/evaluation separation, deter-

ministic seeding, and 99.95% empirical severity monotonicity. Zero validation errors or warnings were recorded across the full release.

2. **Restoration effectiveness follows invertibility, not visual polish.** Degradations with a structured, known forward model — periodic interference, Poisson–Gaussian noise, and (partially) motion blur — are substantially recoverable by matched classical restoration at the pixel level. Degradations that discard information irreversibly — quantization, detector saturation, sampling loss, photobleaching, and JPEG compression — remain information-limited, and contrast-enhancement-based restoration can even reduce measured pixel fidelity while superficially improving visual appearance.

3. **Fine-grained perception is the semantic stress point.** The overall change in VLM macro accuracy under degradation is modest and statistically uncertain at $n = 500$, but this aggregate figure conceals large, family- and severity-specific effects. Strong corruption and fine-perception question types expose meaningful paired failures — and, notably, pixel-level restoration success does not always translate into semantic recovery (e.g. motion blur), while in at least one case (photobleaching) semantic recovery exceeds what pixel metrics alone would predict.

Scope and limitations. All degradations studied here are controlled, *synthetic* corruptions applied to real microscopy images, rather than corruptions captured directly from instrument variability; the Phase 1.5 VLM results are exploratory, based on a single model (Qwen3-VL-8B) and $n = 500$ paired samples, several family-specific subgroups are as small as $n = 19$, and improvement in pixel-level fidelity does not automatically imply improvement in biological or semantic correctness. These results should therefore be read as evidence of feasibility and direction rather than as definitive, model-agnostic conclusions.

We close with the open question that motivated this project: *which image failures should a microscopy-focused vision-language model be trained to notice — and when should it instead admit that the underlying information is simply gone?*

Acknowledgments

This project builds directly on the μ-Bench evaluation benchmark (NeurIPS 2024) and the low-level vision curricula introduced by Q-Instruct (CVPR 2024) and Q-Bench (ICLR 2024). Training and development pool images are drawn from BBBC010, BBBC038, and IDR0067, released under CC0 / CC0-1.0 / CC-BY-4.0 / CC-BY-SA-4.0 licenses.